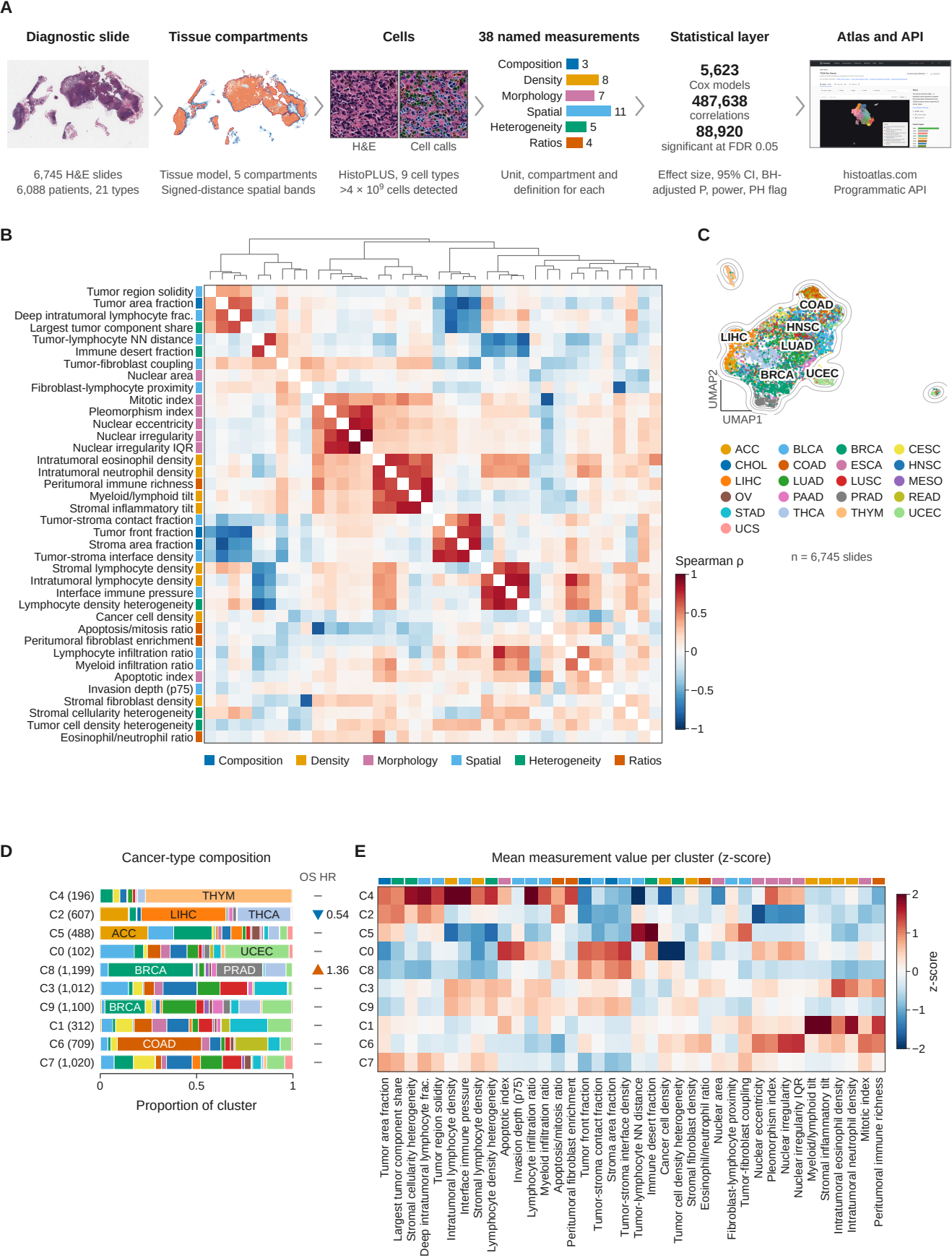

A
Diagnostic slide
Tissue compartments
Cells
38 named measurements
Statistical layer
Atlas and API
H&E
Cell calls
Composition 3
Density 8
Morphology 7
Spatial 11
Heterogeneity 5
Ratios 4
5,623
Cox models
487,638
correlations
88,920
significant at FDR 0.05
6,745 H&E slides
6,088 patients, 21 types
Tissue model, 5 compartments
Signed-distance spatial bands
HistoPLUS, 9 cell types
>4 × 10^9 cells detected
Unit, compartment and definition for each
Effect size, 95% CI, BH-adjusted P, power, PH flag
histoatlas.com
Programmatic API
B
Tumor region solidity
Tumor area fraction
Deep intratumoral lymphocyte frac.
Largest tumor component share
Tumor-lymphocyte NN distance
Immune desert fraction
Tumor-fibroblast coupling
Nuclear area
Fibroblast-lymphocyte proximity
Mitotic index
Pleomorphism index
Nuclear eccentricity
Nuclear irregularity
Nuclear irregularity IQR
Intratumoral eosinophil density
Intratumoral neutrophil density
Peritumoral immune richness
Myeloid/lymphoid tilt
Stromal inflammatory tilt
Tumor-stroma contact fraction
Tumor front fraction
Stroma area fraction
Tumor-stroma interface density
Stromal lymphocyte density
Intratumoral lymphocyte density
Interface immune pressure
Lymphocyte density heterogeneity
Cancer cell density
Apoptosis/mitosis ratio
Peritumoral fibroblast enrichment
Lymphocyte infiltration ratio
Myeloid infiltration ratio
Apoptotic index
Invasion depth (p75)
Stromal fibroblast density
Stromal cellularity heterogeneity
Tumor cell density heterogeneity
Eosinophil/neutrophil ratio
Composition
Density
Morphology
Spatial
Heterogeneity
Ratios
C
UMAP1
UMAP2
COAD
HNSC
LIHC
LUAD
BRCA
UCEC
ACC
BLCA
BRCA
CESC
CHOL
COAD
ESCA
HNSC
LIHC
LUAD
LUSC
MESO
OV
PAAD
PRAD
READ
STAD
THCA
THYM
UCEC
UCS
Spearman ρ
1
0.5
0
−0.5
−1
n = 6,745 slides
D
Cancer-type composition
OS HR
C4 (196)
C2 (607)
C5 (488)
C0 (102)
C8 (1,199)
C3 (1,012)
C9 (1,100)
C1 (312)
C6 (709)
C7 (1,020)
THYM
LIHC
THCA
ACC
UCEC
BRCA
PRAD
BRCA
COAD
0.54
1.36
0
0.5
1
Proportion of cluster
E
Mean measurement value per cluster (z-score)
C4
C2
C5
C0
C8
C3
C9
C1
C6
C7
z-score
2
1
0
−1
−2
Tumor area fraction
Largest tumor component share
Stromal cellularity heterogeneity
Deep intratumoral lymphocyte frac.
Tumor region solidity
Intratumoral lymphocyte density
Interface immune pressure
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